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Revolutionizing Subsurface Evaluation with Digital Transformation e-MARS Vast Enhancement and Upgrades (VENUS): An AI / ML - Driven Advanced Reservoir Management for Field Optimization and Development to Maximize Oil Production

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Abstract

In the upstream sector of the oil and gas industry that manages massive and mature oil fields, optimization in existing wells and new infill well drilling programs have become critical efforts within strategic field development to maximize production and managing revenue performance. The success of the optimization and development programs are highly determined in the evaluation phase to select the most prominent candidates to be executed to deliver the highest value creation. However, these programs pose significant challenges due to complex evaluation processes with sub-optimal results.

Conventionally, the full evaluation process and the underlying individual well-based review are conducted in manual ways relying on human capability, involve long and time-consuming processes, demand significant resources, and require multi-tools. Despite the extensive effort and resources, the conventional method produces inconsistent results, improper prioritization, with inefficient execution and low well performance outcomes.

The implementation of "e-MARS" Vast Enhancement and Upgrades (VENUS) is introduced as a solution. "e-MARS", which stands for electronic-Massive Advance Reservoir Strategy, is a pioneer digital solution for the upstream oil and gas industry using Artificial Intelligence/Machine Learning (AI/ML) technologies and advanced subsurface reservoir management. As a sustainable innovation, it was developed in stages, starting by the predecessor "e-MARS" as the pilot project for idle well reactivation evaluation. Then, the version is expanded to VENUS (Vast Enhancement and Upgrade) as the next stage of "e-MARS" which enhanced analytical capabilities to cover all existing wells and new infill well drilling. It successfully transforms the traditionally human-led, manual, and individual well-based evaluation process into a robust, comprehensive, efficient, automated digital solution through AI/ML driven Advanced Reservoir Management, bringing in-depth and comprehensive analysis which encompasses various analytical procedures that perform simultaneous, real-time, high-resolution evaluation with technical precision and consistency, leading to strategic prioritization and high-grading execution.

"e-MARS" (VENUS) marks the new era of field development evaluation process (optimization in existing and new infill wells drilling programs) as a robust digital solution. This method has been successfully implemented in over 3,500 wells (covering 4 fields and 15 productive reservoirs) in Southeast Asia's largest mature waterflood field and widely replicated to other fields in Indonesia. The effective implementation of "e-MARS" delivered substantial impacts, contributing significantly to national oil production, setting a technical benchmark for future field optimization and development initiatives, as well as lays the foundation for unlocking opportunities across the region.

Background

Field development activities (optimization in existing and new infill wells drilling programs) are essential for extending field life in the upstream sector of the oil and gas industry to maximize production and managing revenue performance from mature oil fields. The success of optimization and development programs is strongly influenced by the quality of the evaluation phase to pinpoint high-potential candidates to be executed for maximizing revenue. However, the intricate nature and complexity of the evaluation process, starting from data gathering, subsurface analysis including remaining oil identification, completion strategy, production performance forecasting and execution prioritization becomes a challenge and hinders process effectiveness, leading to suboptimal outcomes.

The adoption of new approach and innovative technology is required to address the problem by integrating technical expertise in subsurface reservoir management with advanced analytics using artificial intelligence (AI) as a key enabler. Artificial Intelligence technology plays a crucial role in enabling multifaceted subsurface evaluations, including complex reservoir modeling, simulations and opportunity identification to increase oil production.

Objectives & Scope

This paper introduces new method in field development evaluation process through "e-MARS" (VENUS), a robust digital solution that has been successfully implemented in over 3,500 wells in Southeast Asia's largest mature waterflood field and widely replicated to other fields in Indonesia.

This paper covers conceptual foundation, methodologies, technological framework and key impact areas of solution, providing a comprehensive view of conventional challenges in optimization and development programs, Advanced Reservoir Management strategy, Artificial Intelligence utilization for advance analytics, successful deployment in large-scale mature fields, including key performance metrics and value creation.

Challenges Applying Conventional Method in Field Development

An oil and gas company managing large-scale and mature fields faces inherent challenges in achieving production targets due to complex operations and uncertainty in subsurface conditions. Such challenges are evident in Field X, the largest waterflood field in Southeast Asia with thousands of wells that have been produced for over seven decades. Sustaining production in this field has become increasingly challenging due to continuous decline. Extensive efforts, including massive optimization programs in existing and new infill wells drilling have been implemented to increase production. However, the programs face several persistent issues, such as:

- Sub-optimum results due to narrow evaluation method (individual well-based evaluation)
- Manual processes, where evaluations are performed manually by practitioners
- Long process due to complex evaluation which requires a significant amount of time

- Inconsistent result due to varying level of competencies among practitioners
- Multi-tools utilization with numerous tools required for multifaceted review and data retrieval from various sources
- Lack of prioritization with first come first serve approach in execution
- Inefficient execution due to longer duration and high cost

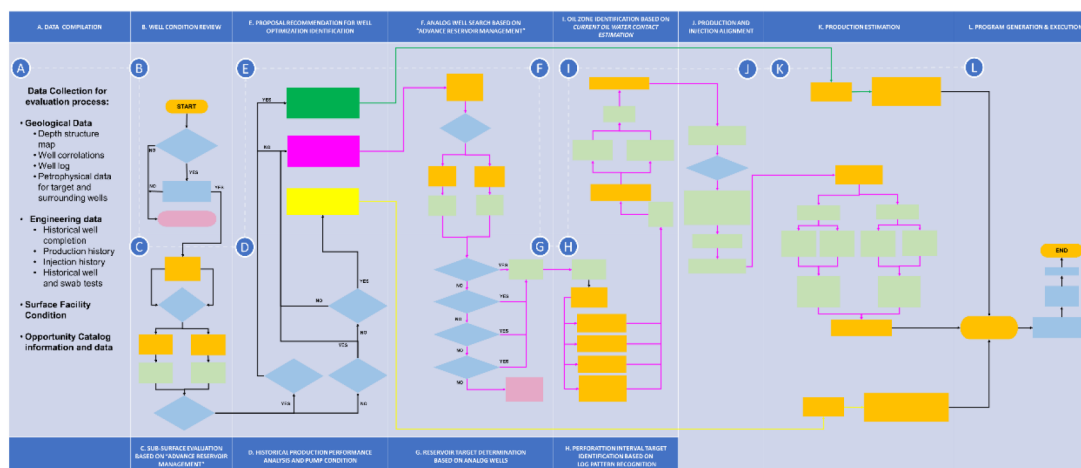


Figure 1—Illustration Long Conventional Evaluation Process

Individual Well Review



Multiple Application

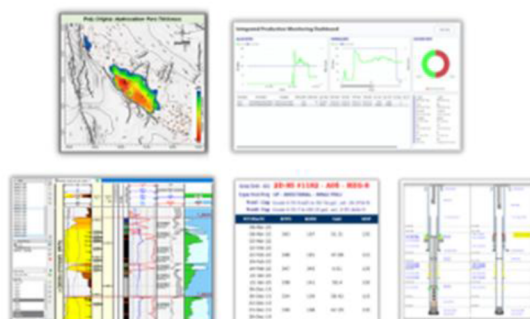


Figure 2—Illustration individual well-based evaluation

Methodology

The implementation of "e-MARS" (VENUS) is introduced as a robust and comprehensive digital solution to address the existing challenges. This method was initially developed and implemented in Field X, the largest waterflood field in Southeast Asia with thousands of wells that have been produced for over seven decades. It successfully transforms the traditionally human-led, manual individual well-based evaluation process into a robust, comprehensive, efficient, automated digital solution through AI/ML driven Advanced Reservoir Management, bringing in-depth and comprehensive analysis which encompasses various analytical procedures that perform simultaneous, real-time, high-resolution evaluation with technical precision and consistency, leading to strategic prioritization and high-grading execution.

Advanced Reservoir Management Strategy

Field X is a giant oil field with a complex geological history characterized by a mix of fluvial and deltaic depositional systems. Structurally, the field comprises multiple closures influenced by thrust faults which have played a role in the structural evolution of the basin and hydrocarbon migration. The field that has been in production for over seven decades and waterflooded since 1970s. Majority of production and waterflood injections targeting high quality reservoir (HQR) that have been breakthrough and currently produce with high water cut (WC).

Historically, field development evaluations were performed on individual well based reviews which focused on high-quality reservoirs (HQR) as primary target. As a result, the programs have yielded suboptimal performance.

Emerging insights from recent studies and pilot projects highlight significant potential in lower-quality reservoir (LQR) zones that have remained underexploited both in crestal and flank area of the field, due to poor sweeping efficiency caused by permeability contrast. Whereas the HQR showing good remaining oil concentrated at crestal and depleted at mid-flank area. These new insights have led to the formulation of an Advanced Reservoir Management concept to redefine field development strategies.

Advanced Reservoir Management is an adaptive and dynamic process, with one of its core principles emphasizing the strategic management of production zones through in-depth evaluation and identification of reservoir characteristics and system behavior. Insights from the studies and pilot project, along with well analog data, form the basis of the Advanced Reservoir Management strategy, which adopts a "back to geology" approach which focused on three key aspects:

- Structural geological closure: well locations in relation to subsurface structural depth, distinguishing between crestal (up-dip structure) and flank (down-dip structure) area
- Reservoir characteristics: classification of reservoir zones based on rock properties, which indicate distinct behavior in relation to ongoing production and injection activities, categorized into High Quality Reservoirs (HQR) with blocky sand patterns and Lower Quality Reservoirs (LQR) with fining-upward sequences, shaly/sandwich sands, and thin sand layers.
- Injection-production alignment: strategic synchronization of injection and production to maximize reservoir sweep and minimize water cycling.

In general, the Advanced Reservoir Management evaluation framework which incorporates geological aspects as key parameters has been proven to enhance oil production. The framework consists of two primary strategies:

1. High Quality Reservoirs (HQR), characterized by high permeability, are targeted for production exclusively in crestal (up-dip) areas, where significant remaining oil potential exists. These zones typically benefit from upward oil migration driven by peripheral water injections.
2. Lower Quality Reservoirs (LQR), defined by lower permeability, are selected for development in both mid-crestal and flank (down-dip) areas which are supported by nearby injections. In these reservoirs, remaining oil is often trapped due to permeability contrast, resulting in reduced sweeping efficiency and limited production performance.

The Advanced Reservoir Management strategy as fundamental reference for field development evaluation, shifting the conventional evaluation method from individual well based review to a comprehensive assessment encompasses various complex analytical procedures, including structural segmentation; production-injection alignment; Current Oil Water Contact (COWC) level estimation, reservoir target determination (perforation interval detection through log pattern recognition algorithm), prognosis/synthetic well log data that represent current subsurface reservoir condition, development

recommendation proposal (for existing wells optimization and new infill wells drilling location point), high grading analysis, reserve estimation and production forecasting for economic assessment.

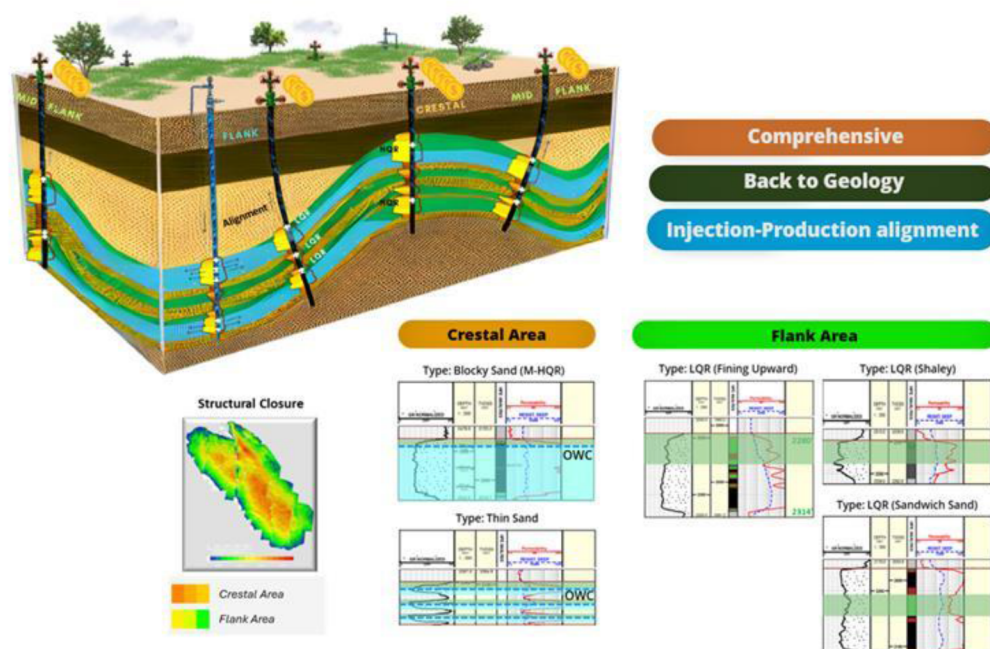


Figure 3—Illustration of Advance Reservoir Management Strategy

Advanced Analytics Using Artificial Intelligence (AI)

Artificial intelligence (AI) technology has rapidly emerged as a key enabler in today's oil and gas industry. It plays a critical role in facilitating multifaceted subsurface evaluations—supporting complex reservoir modeling, simulation, and opportunity identification to enhance oil production. In the context of field development evaluation, AI has been adopted to assist, enhance, and accelerate multifaceted and comprehensive assessments. A hybrid AI system is deployed, combining multiple artificial intelligence techniques to tackle complex evaluation and analysis in simultaneous, real-time, continues and consistent manner with technical precision enabling strategic prioritization and high-grading execution decisions.

AI Expert system is applied for evaluating optimization opportunities in existing wells with proven Advance Reservoir Management strategy as knowledge base. The solution shifts the evaluation process from a conventional manual, human-dependent, complex and time-consuming reviews into a fully automated, advanced analytical workflow that ensures technical robustness and consistency with exponentially faster and more efficient. The AI-driven evaluation has been streamlined from the manual repetitive 30-steps workflow, performed in an iterative well-by-well evaluation process into a single automated step, end to end digital solution that seamlessly performing sequence of assessment and unifying stages of evaluation process simultaneously for thousands of wells, encompassing following main processes:

- Data gathering and processing: covering automatic and real time data retrieval of both dynamic and static field datasets, covering subsurface (e.g., geological, petrophysical, reservoir, production data and wellbore configuration) and surface-level information (e.g., surface facility readiness), ensuring a holistic and accurate foundation for advanced analysis.
- Structural closure segmentation: structural segments are defined using water shed algorithm referring to one of advanced reservoir management concepts to include geological closure interpretations, enabling targeted analysis aligned with reservoir compartmentalization. This

structural framework guides subsequent evaluations, ensuring that each area is assessed in context of its unique geologic behavior and production potential.

- Reservoir target selection: algorithm reservoir targets are determined through data-driven interpretation incorporating advanced reservoir management strategy with key geological, petrophysical, and production characteristics.
- Estimation of Current-Water Contact (OWC): The evaluation includes a dynamic estimation of the current oil-water contact level within the reservoir, integrating historical production data and log-derived fluid contacts. This insight is essential for identifying upswept zones and enhancing accuracy in target selection.
- Analog well identification for reference: strategic identification of analog wells is performed to serve as reference points for evaluation. These wells, selected based on similar reservoir characteristics, structural positioning, and production behavior, provide empirical benchmarks to enhance predictive accuracy and validate assumptions.
- Perforation interval determination through log pattern recognition: advanced algorithms are employed to analyze log signatures and identify optimal perforation intervals based on characteristic reservoir responses. This data-driven approach leverages pattern recognition techniques to detect productive intervals with high accuracy to maximize hydrocarbon efficiency. The defined perforation intervals are prioritized based on analog wells data.
- Injection–production alignment assessment: the evaluation is conducted to align injection patterns with targeted produced reservoir intervals, minimizing water cycling. This process integrates dynamic reservoir behavior, injection conformance, and pressure support analysis to optimize fluid movement and enhance hydrocarbon recovery across structurally and stratigraphically defined zones.
- Production forecasting: a predictive fundamental approach is conducted to estimate oil and fluid production. This involves integrating historical performance data, structural closure and reservoir characteristics to forecast production outcomes with technical confidence—supporting economic justification and execution prioritization
- Formulation of final program recommendation: holistic integration of all evaluation components is performed to generate the final recommendation. This proposal integrates subsurface insights, production forecasts, economic indicators, and technical feasibility assessments assuring that each selected candidate is prioritized based on impact potential, execution readiness, and value creation for optimized field development planning.

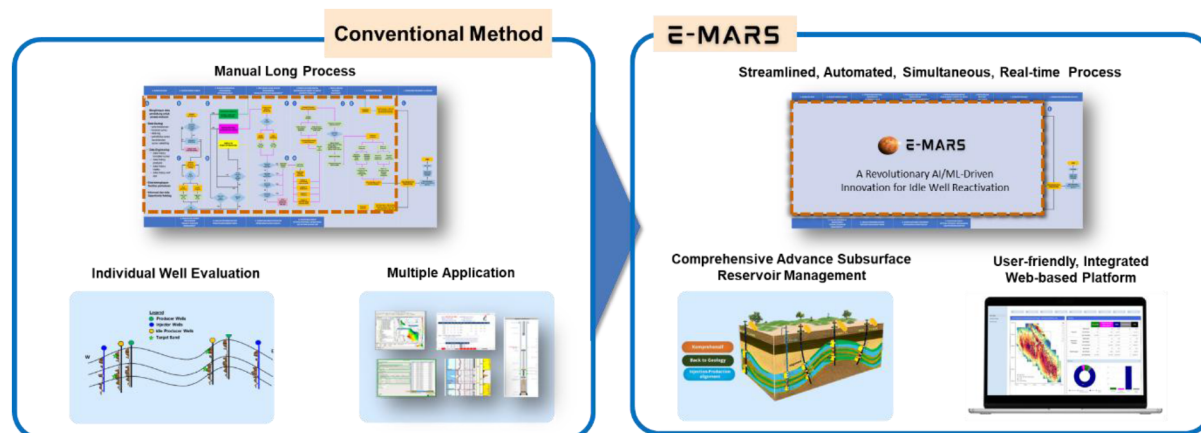


Figure 4—Illustration of process comparison between conventional method and "e-MARS" (VENUS)

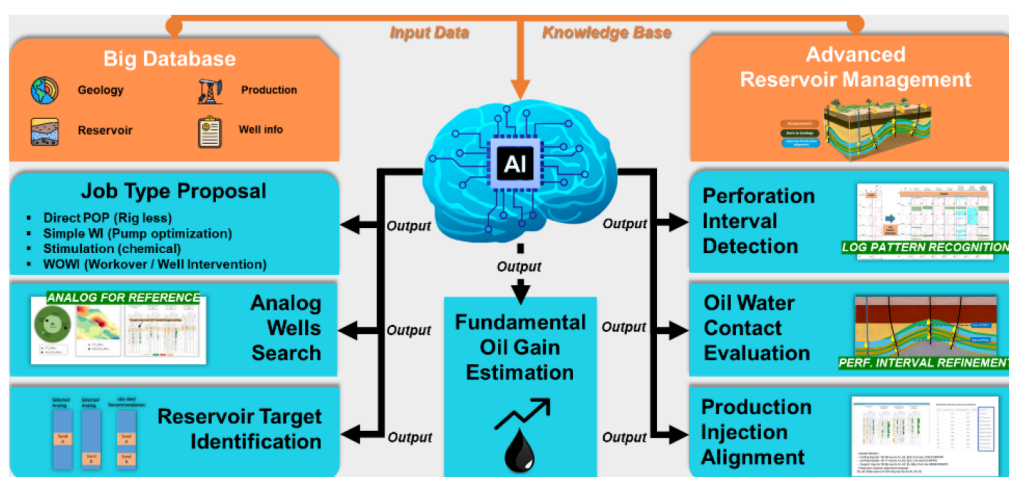


Figure 5—Illustration AI Driven Advance Reservoir Management Process

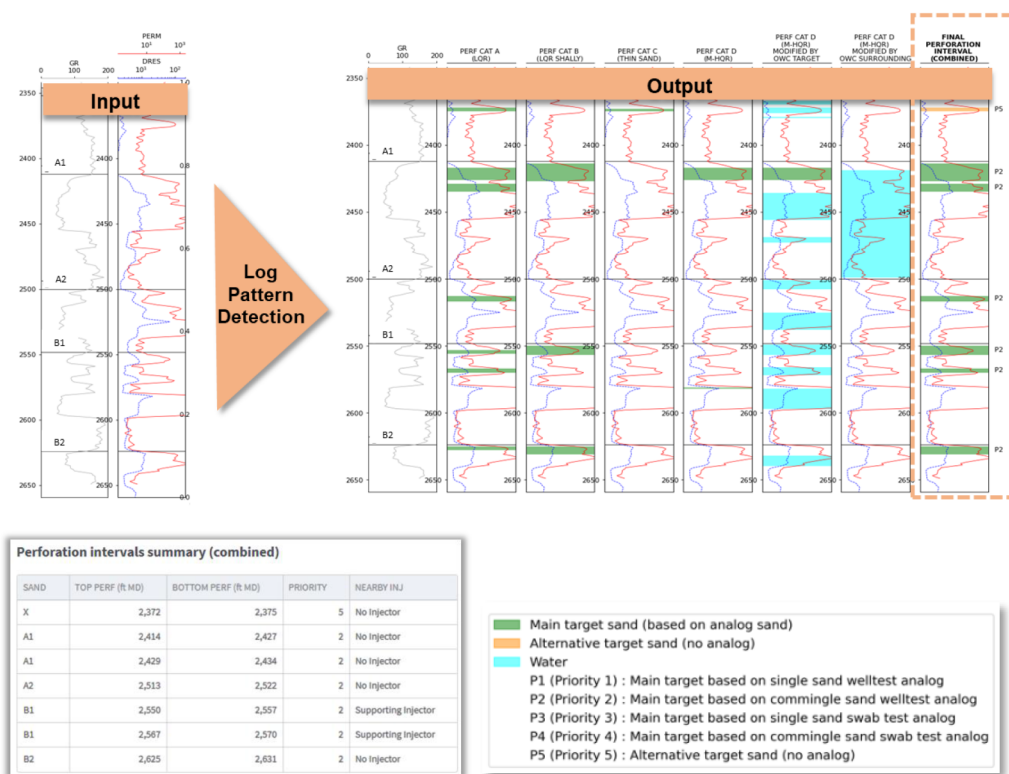


Figure 6—Illustration sample feature - Log Pattern Recognition Algorithm for Production Interval Detection

Building on the same framework used for existing wells optimization evaluation process, "e-MARS" (VENUS) transforms the new infill well drilling evaluation process from traditional, manual workflows into in-depth advanced automated process using hybrid AI algorithm that primarily driven by machine learning (Supervised MLP), complemented by AI Expert system with Advance Reservoir Management as knowledge base and reasoning techniques using fuzzy logic method.

The new infill well drilling evaluation process is conceptually structured around a series of integrated technical assessments. Each stage contributes to a comprehensive understanding of subsurface potential and development feasibility. The workflow encompasses the following key evaluation components:

1. Pin-point the new infill well locations: the process involves determining the most optimal subsurface coordinates for new infill wells drilling through a reduced spacing strategy, utilizing Voronoi diagram.

This method delineates the reservoir into discrete drainage areas based on existing well configurations, allowing precise detection of under-swept and under-drained zones. By analyzing inter-well distances, drainage radius, and areal reservoir distribution, the process enhances sweep efficiency and informs well placement strategies aimed at maximizing hydrocarbon recovery and development effectiveness.

2. Subsurface reservoir condition prediction: the evaluation process includes generating prognosis or synthetic well log data that represent the current subsurface conditions at the proposed drilling location. Machine-learning approaches are utilized to develop derivative well log value predictive model based on basic triple combo log value. This model is trained using a supervised learning approach with Multilayer Perceptron (MLP) regressor. To predict well log value in prospect well, basic triple combo value is calculated by applying 3D-spatial interpolation using Radial Basis Function (RBF) as input to derivative log predictive model. This predictive approach leverages spatial correlation with surrounding well data and geological trends to simulate lithological and petrophysical properties, enabling informed reservoir characterization and reducing drilling uncertainties. This approach offers a promising alternative to substitute the requirement for conventional surveillance well logging by generating synthetic well data, enabling real-time reservoir evaluation and target identification with reduced operational cost.
3. Production interval target determination: the determination of potential reservoir production intervals in proposed infill wells is carried out using an AI-based expert system with Advance Reservoir Management as knowledge base incorporated in existing well optimization evaluation process. Advanced log pattern recognition algorithms are applied to predict and delineate the optimal perforation intervals, enabling precision in reservoir interval perforation and enhancing productivity. Identification of analog wells with comparable geological and production attributes is also performed to provide reliable reference models and benchmarks.
4. New infill well proposal prioritization: a fuzzy logic-based high-grading system is utilized to evaluate and prioritize infill drilling proposals by integrating technical viability. This ranking method accounts for uncertainty in subsurface and production parameters, enabling a more objective and robust selection of the most value-driven candidates for execution.
5. Economic assessment: economic evaluation of new infill well proposals is conducted using fundamental approach and data-driven methodology that integrates historical production performance, structural closure analysis, and reservoir characterization. This predictive framework enables technically confident estimation of oil and fluid production, as well as recoverable reserves. The resulting forecasts serve as a foundation for economic justification and strategic execution prioritization, ensuring value-driven decision-making underpinned by robust subsurface understanding.

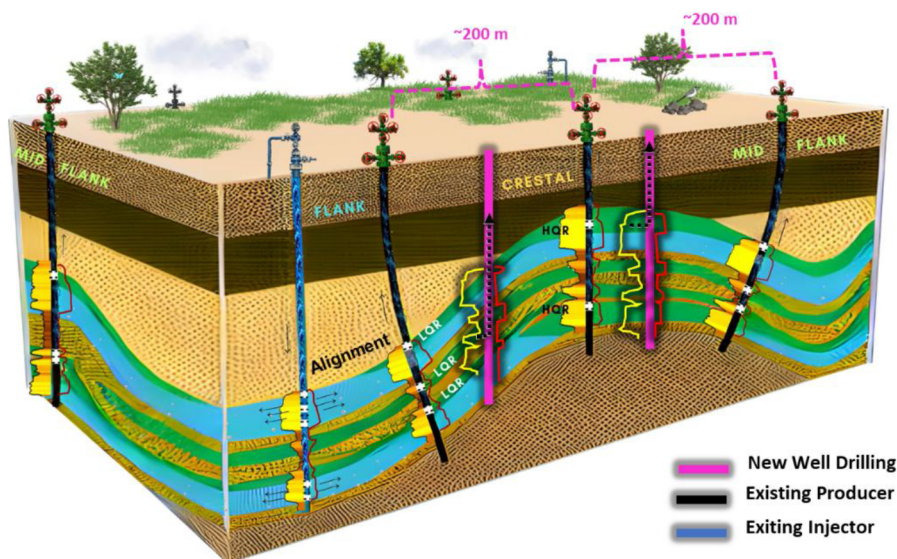


Figure 7—Illustration of new infill wells drilling evaluation concept

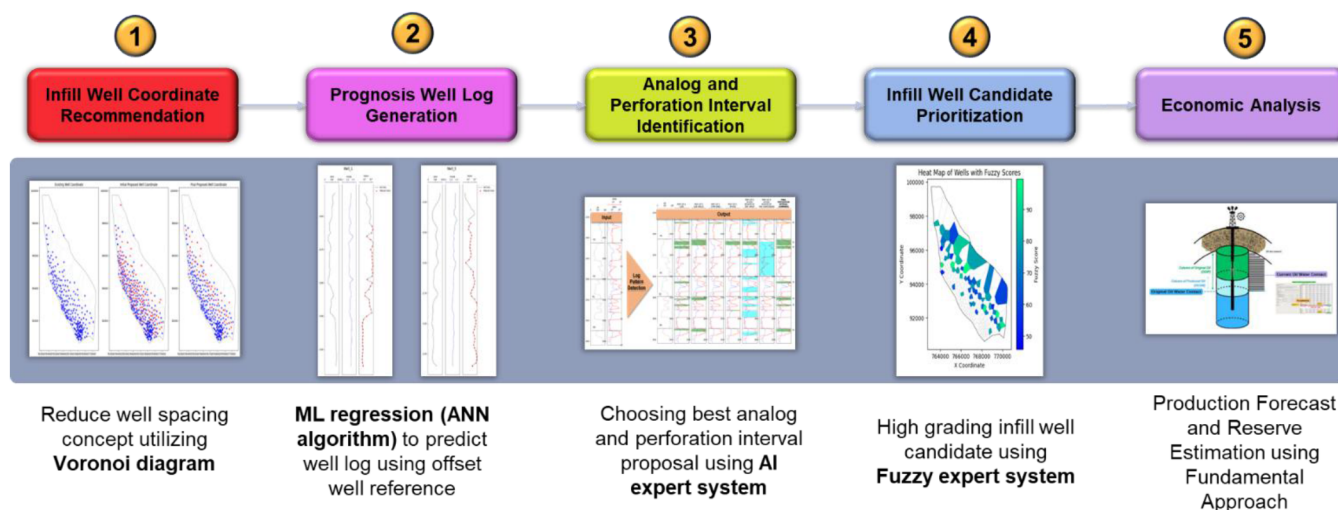


Figure 7—New infill well drilling evaluation process using "e-MARS" (VENUS)

Results

"e-MARS" (VENUS) has been fully deployed for more than 3,500 wells evaluation in Southeast Asia's largest mature waterflood field and widely replicated to other fields in Indonesia. The effective implementation of "e-MARS" delivered substantial impacts, contributing significantly to national oil production, setting a technical benchmark for future field optimization and development initiatives, as well as lays the foundation for unlocking opportunities across the region.

This method delivers outstanding impacts in comparison to the conventional method and other best practices in the world, by doubling the success rate and oil production as well as exponentially accelerating evaluation with actual value creation \$100 Million, equivalent with 1.3 MMBO. Furthermore, "e-MARS" has delivered cost savings of \$20 million through its in-house, on-premises, avoiding the need for expensive external solutions.

Beyond operational efficiency, "e-MARS" supports broader sustainability and socio-economic goals. By optimizing well selection and execution strategies, it helps reduce unnecessary operations that contribute to wastewater and carbon emissions, promoting environmental stewardship. Additionally, the success of the

optimization programs has a positive ripple effect on local economies, particularly in regions where these programs are implemented, by generating job opportunities and boosting local commerce.

Conclusion

"e-MARS" (VENUS) utilized a hybrid AI system, combining multiple artificial intelligence techniques to conduct complex evaluation and advance analysis of field development programs which transformed the conventional human-led, manual individual well-based evaluation process into a robust, comprehensive, efficient, automated digital solution through AI/ML driven Advanced Reservoir Management. This method brings in-depth and comprehensive analysis which encompasses various analytical procedures that perform simultaneous, real-time, consistent and exponentially faster evaluation with technical robustness, enabling strategic prioritization and high-grading execution.

This method delivers impactful results compared to the conventional method and other best practices in the world, by doubling the success rate and oil production in field development programs, leading to maximum value generation

"e-MARS" (VENUS) is highly adaptable for replication in other oil and gas companies, as it requires standard infrastructure, making it a feasible solution for organizations seeking to continue to strive for technological capabilities advancement using AI/ML Technology to enable multifaceted evaluations, adopt cutting-edge technology without the need for specialized or proprietary systems.

Novelty

"e-MARS" (VENUS) marks the new era of field development evaluation process (optimization in existing and new infill wells drilling programs) as a robust digital solution. It reflects high commitment for technology adoption aimed for digitalization and efficiency to maximize revenue.

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